Code No: 123BP **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, April/May - 2018
DATA STRUCTURES
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is space complexity? [2]
b) What is Big-O and Omega Notation? [3]
c) What is a Queue? Give its basis operation. [2]
d) Convert to postfix: 2+3*4+6/2. [3]
e) State the different representations of a Graph. [2]
f) What is a Max-heap? [3]
g) What is the time complexity of linear search? [2]
h) What is a hash function? Give example. [3]
i) What is a Red-Black Tree? [2]
j) What is an AVL Tree? [3]

PART-B

(50 Marks)

2. What is a Circular Linked List? Write an algorithm for insertion and searching an element into a circular linked list. [10]

OR

3.a) How to represent One, Two, Three and Multidimensional arrays in C? Explain with example.
b) What is a sparse matrix? Give representations. [5+5]

4. What is a Stack? Explain all the primitive operations performed on a stack. [10]

OR

5. What is a Circular Queue? Explain various operations on a Circular Queue. [10]

6.a) Construct an expression tree for the given expression: A+(B-C)*D+(E*F).
b) A binary tree has 9 nodes. The inorder and postorder traversal sequences are given below. Give the preorder traversal.
Inorder: D G B A H E C F I
Postoder: G D B H E I F C A [5+5]

OR

7. Explain BFS and DFS algorithms. [10]

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8. Explain linear search and binary search. Which method is efficient in terms of complexity? [10]

OR

9. Explain how to sort the array of elements using heap sort. [10]

10. Write an algorithm for insertion and deletion in a Binary Search Tree? [10]

OR

11. Show each step of the AVL Tree built from a sequence of insertions corresponding to the following keys 44,17,32,78,50,54,62,48,88? [10]

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